Light

GCSE Computer Science - Cyber Security Prevention (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What are biometric security measures based on?

- **A.** Something you know, such as a password
- **B.** Something you have, such as a key card
- **C.** Unique physical or behavioural characteristics of a person
- **D.** The time of day you try to log in

Q2 (1 mark)

Why are fingerprint scanners commonly used on mobile devices?

- **A.** They make the phone lighter

- ○ **B.** They are quick, convenient, and provide strong authentication without requiring users to remember passwords
- ○ **C.** They improve the phone's battery life
- ○ **D.** They make the screen brighter

Q3 (1 mark)

What is the primary purpose of CAPTCHA?

- ○ **A.** To distinguish between human users and automated programs (bots)
- ○ **B.** To encrypt data sent over the internet
- ○ **C.** To speed up website loading times
- ○ **D.** To compress images on web pages

Q4 (1 mark)

Which characteristic makes a password stronger?

- ○ **A.** Using your name and birthday
- ○ **B.** Using the same password everywhere
- ○ **C.** Being short and simple to type
- ○ **D.** Including a mix of uppercase, lowercase, numbers, and symbols

Q5 (1 mark)

Why do websites send email confirmations when you create a new account?

- ○ **A.** To advertise their products
- ○ **B.** To verify that you own and can access the email address you provided
- ○ **C.** To slow down the registration process
- ○ **D.** To fill up your inbox

Q6 (1 mark)

What is the main security benefit of automatic software updates?

- ○ **A.** They install security patches that fix known vulnerabilities as soon as they become available
- ○ **B.** They make your computer screen larger
- ○ **C.** They add new games to your computer
- ○ **D.** They delete all your files

Q7 (1 mark)

How does facial recognition unlock a mobile phone?

- ○ **A.** By recognising your voice
- ○ **B.** By scanning your fingerprint

- ○ **C.** By analysing the unique structure and features of your face
- ○ **D.** By reading your thoughts

Q8 (1 mark)

What type of software helps protect against viruses, trojans, and spyware?

- ○ **A.** Anti-malware or antivirus software
- ○ **B.** Word processing software
- ○ **C.** Spreadsheet software
- ○ **D.** Video editing software

Q9 (1 mark)

What is the best protection against social engineering attacks?

- ○ **A.** Installing faster internet
- ○ **B.** Using a bigger monitor
- ○ **C.** Buying a more expensive computer
- ○ **D.** User awareness and verification of requests through trusted channels

Q10 (1 mark)

What does two-factor authentication (2FA) require?

- ○ **A.** Two different types of proof of identity, such as a password and a code sent to your phone
- ○ **B.** Two people to log in at the same time
- ○ **C.** Two different passwords
- ○ **D.** Logging in twice in a row

Q11 (1 mark)

What does a firewall do?

- ○ **A.** Prevents your computer from overheating
- ○ **B.** Cleans dust from inside your computer
- ○ **C.** Makes your internet connection faster
- ○ **D.** Monitors and controls incoming and outgoing network traffic based on security rules

Q12 (1 mark)

Why is iris scanning considered highly accurate for biometric authentication?

- ○ **A.** Because everyone's iris changes every day
- ○ **B.** Because iris patterns are easy to copy
- ○ **C.** Because iris scanners are the cheapest option

○ **D.** Because the iris pattern is unique to each individual and remains stable throughout life

Q13 (2 marks)

A website uses a CAPTCHA that shows nine images and asks users to 'select all images containing traffic lights'. Why is this type of challenge effective?

○ **A.** Because humans cannot recognise traffic lights

○ **B.** Because computers can solve this faster than humans

○ **C.** Because the task requires visual recognition and contextual understanding that humans perform easily but current AI systems find challenging, especially with varied image angles and partial views

○ **D.** Because traffic lights are found in every country

Q14 (2 marks)

A company implements a password policy requiring passwords to be at least 12 characters, include mixed case and numbers, and be changed every 90 days. Why might enforced regular password changes actually reduce security in some cases?

○ **A.** Because frequent mandatory changes often lead users to choose weaker, more predictable passwords or write them down, potentially creating greater vulnerabilities than longer-lasting strong passwords

○ **B.** Because longer passwords are less secure

○ **C.** Because computers cannot handle password changes

○ **D.** Because 90 days is too long between changes

Q15 (2 marks)

When you request a password reset, why do legitimate services send a time-limited link to your email rather than simply displaying your current password?

- ○ **A.** To make the process take longer
- ○ **B.** Because passwords should be stored as hashes that cannot be reversed, and time-limited links verify email ownership while preventing link reuse if intercepted
- ○ **C.** Because sending passwords by email is faster
- ○ **D.** Because websites do not actually know your password

Q16 (2 marks)

What is a significant limitation of using biometrics compared to passwords for authentication?

- ○ **A.** Biometrics are too easy to guess
- ○ **B.** Biometrics require internet connection at all times
- ○ **C.** Passwords are more unique than fingerprints
- ○ **D.** Biometrics cannot be changed if compromised - you cannot get new fingerprints, but you can always create a new password

Q17 (2 marks)

Why might a large organisation choose to delay automatic updates for a short period rather than applying them immediately?

- ○ **A.** Because updates are always harmful

- ○ **B.** To test updates in a controlled environment first, ensuring they do not break critical business applications before deployment across the organisation
- ○ **C.** Because older software is always more secure
- ○ **D.** To save money by using fewer updates

Q18 (2 marks)

A user receives an email with a link that appears to go to 'www.mybank.com' but actually leads to 'www.myb4nk.com'. What security practice would help identify this deception?

- ○ **A.** Hovering over links to preview their actual destination URL before clicking, and navigating to websites by typing addresses directly rather than clicking email links
- ○ **B.** Reading emails faster
- ○ **C.** Using a smaller font size
- ○ **D.** Opening all email attachments immediately

Q19 (2 marks)

Why is SMS-based two-factor authentication considered less secure than authenticator apps?

- ○ **A.** Because SMS messages are too fast
- ○ **B.** Because SMS can be intercepted through SIM swapping attacks or network vulnerabilities, while authenticator apps generate codes locally on your device without network transmission
- ○ **C.** Because authenticator apps are more expensive

- ○ **D.** Because SMS only works during daytime

Q20 (2 marks)

A home user notices their firewall blocked an incoming connection attempt on port 3389 from an unknown IP address. What does this indicate?

- ○ **A.** The firewall detected and blocked a potential unauthorised attempt to connect to the computer's Remote Desktop service, demonstrating the firewall working as intended
- ○ **B.** The firewall is broken and needs replacing
- ○ **C.** The user's internet is not working properly
- ○ **D.** Port 3389 should always be open for safety

Q21 (2 marks)

What is the difference between real-time protection and scheduled scans in anti-malware software?

- ○ **A.** They are exactly the same thing with different names
- ○ **B.** Real-time protection only works when the computer is turned off
- ○ **C.** Scheduled scans are faster than real-time protection
- ○ **D.** Real-time protection monitors files and activities continuously as they happen, while scheduled scans periodically check all stored files - both are needed for comprehensive protection

Q22 (2 marks)

An employee receives a phone call from someone claiming to be IT support, asking them to download remote access software. What verification step should the employee take?

- ○ **A.** Download the software immediately to avoid keeping IT waiting
- ○ **B.** Ask the caller for their employee ID number and then trust them
- ○ **C.** End the call and contact the IT department using contact details from the company directory, not any number provided by the caller
- ○ **D.** Give the caller their password so they can verify they are from IT

Q23 (2 marks)

What is the trade-off that biometric systems must balance between false acceptance rate and false rejection rate?

- ○ **A.** There is no trade-off - both can be zero simultaneously
- ○ **B.** Making the system more strict reduces false acceptances (impostors getting through) but increases false rejections (legitimate users being denied), and vice versa
- ○ **C.** Higher false acceptance makes the system faster
- ○ **D.** False rejection rate only affects computers, not people

Q24 (3 marks)

A bank's online system uses passwords, SMS verification codes, CAPTCHA on login attempts, automatic account lockout after failed attempts, and encrypted connections. Why does the bank use multiple security measures rather than relying on just one very strong measure?

- ○ **A.** Because using more measures makes the website look more professional

○ **B.** To waste customers' time and discourage them from banking online

○ **C.** Defence in depth ensures that if one security layer fails or is bypassed, other layers continue to provide protection - no single measure is perfect, so multiple layers create overlapping protection that is harder for attackers to fully penetrate

○ **D.** Because the bank could not decide which single measure was best

Q25 (3 marks)

A company implements these security measures: (1) Biometric fingerprint scanners for building access; (2) Unique password-protected accounts for each employee; (3) Automatic software updates on all computers; (4) Regular security awareness training. Analyse how these measures protect against different types of threats.

○ **A.** These measures all protect against the same single threat

○ **B.** Only the fingerprint scanners provide any real security

○ **C.** Biometric access controls physical entry preventing unauthorised building access; individual accounts provide accountability and limit damage from compromised credentials; automatic updates close software vulnerabilities before exploitation; training protects against social engineering that bypasses technical controls - each addresses a different threat type

○ **D.** Security awareness training is unnecessary if technical measures are in place

Q26 (3 marks)

A website requires email confirmation for: (1) New account registration; (2) Password reset requests; (3) Changing the email address on an existing account. Analyse how these confirmation requirements protect against different attack scenarios.

- ○ **A.** Email confirmations only prevent typing mistakes
- ○ **B.** Email confirmations slow down legitimate users without security benefit
- ○ **C.** Only password reset confirmation provides security
- ○ **D.** Registration confirmation prevents fake accounts using others' email addresses; password reset confirmation prevents attackers who do not control the victim's email from taking over accounts; email change confirmation prevents attackers who have temporarily accessed an account from permanently hijacking it by changing the recovery email

Q27 (3 marks)

A school network has these issues: (1) Teachers share one generic 'teacher' account for the student records system; (2) The antivirus software license expired six months ago; (3) Students have discovered the WiFi password is written on a whiteboard in the staff room. Analyse the security implications.

- ○ **A.** These are minor issues that do not significantly affect security
- ○ **B.** Shared accounts eliminate individual accountability making it impossible to trace data access or modifications; expired antivirus leaves systems vulnerable to known malware threats; disclosed WiFi credentials allow students to access staff network resources potentially reaching sensitive student records - collectively exposing student data to significant risk
- ○ **C.** Only the expired antivirus is a real concern
- ○ **D.** Schools are not targets for cyber attacks so these issues do not matter

Q28 (3 marks)

A hospital implements fingerprint scanners to control access to medication storage areas. The system logs which staff member accessed which cabinet at what time. Analyse the security and accountability benefits this provides.

- ○ **A.** Biometric access ensures only authorised staff can access controlled medications preventing theft; the audit log creates accountability tracking exactly who accessed what and when for compliance and investigation purposes; unlike keys or PIN codes, fingerprints cannot be shared or stolen enabling positive identification of the individual who accessed medications
- ○ **B.** Fingerprint scanners only save time compared to keys
- ○ **C.** The audit log is unnecessary if fingerprint access is required
- ○ **D.** Keys would provide exactly the same security benefits

Q29 (3 marks)

In 2017, the WannaCry attack affected thousands of organisations worldwide, including parts of the NHS. The attack exploited a Windows vulnerability for which Microsoft had released a patch two months earlier. Analyse what this incident reveals about update management.

- ○ **A.** Patches are not important because attacks happen anyway
- ○ **B.** Organisations that applied the patch were protected; those that delayed patching were vulnerable - demonstrating that the window between patch release and patch application is critical, that attackers actively target known vulnerabilities, and that update policies must balance testing needs against the real cost of delayed protection
- ○ **C.** Microsoft should not have released the patch
- ○ **D.** The NHS should have used different software instead of Windows

Q30 (3 marks)

Password managers store all passwords in an encrypted vault protected by one master password. Some people argue this creates a 'single point of failure' - if the master password is compromised, all accounts are exposed. Analyse this concern and the counter-arguments.

- ○ **A.** Password managers should never be used because of this risk
- ○ **B.** Master passwords cannot be compromised so there is no real concern
- ○ **C.** The single point of failure means password managers provide no security benefit
- ○ **D.** The concern is valid but must be weighed against alternatives: without a manager, people reuse weak passwords across sites - one breach exposes all accounts anyway; password managers enable unique strong passwords for every site; the master password can be very strong since it is the only one to remember; managers offer additional protections like encryption and breach monitoring - making them more secure overall despite the theoretical single-point concern

Total: **55** marks • Time: **30 min**

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